

Please subscribe to the official Codeforces channel in Telegram via the link https://t.me/codeforces_official.

PROBLEMS

SUBMIT

STATUS

STANDINGS

CUSTOM TEST

D. Powerful array

time limit per test: 5 seconds

memory limit per test: 256 megabytes

An array of positive integers $a_1, a_2, \dots, a_n$ is given. Let us consider its arbitrary subarray $a_l, a_{l+1}, \dots, a_r$, where $1 \leq l \leq r \leq n$. For every positive integer s denote by K_s the number of occurrences of s into the subarray. We call the *power* of the subarray the sum of products $K_s \cdot K_s \cdot s$ for every positive integer s . The sum contains only finite number of nonzero summands as the number of different values in the array is indeed finite.

You should calculate the power of t given subarrays.

Input

First line contains two integers n and t ($1 \leq n, t \leq 200000$) — the array length and the number of queries correspondingly.

Second line contains n positive integers a_i ($1 \leq a_i \leq 10^6$) — the elements of the array.

Next t lines contain two positive integers l, r ($1 \leq l \leq r \leq n$) each — the indices of the left and the right ends of the corresponding subarray.

Output

Output t lines, the i -th line of the output should contain single positive integer — the power of the i -th query subarray.

Please, do not use `%lld` specifier to read or write 64-bit integers in C++. It is preferred to use `cout` stream (also you may use `%I64d`).

Examples

input	Copy
3 2 1 2 1 1 2 1 3	
output	Copy
3 6	

input	Copy
8 3 1 1 2 2 1 3 1 1 2 7 1 6 2 7	
output	Copy
20 20 20	

Note

Consider the following array (see the second sample) and its [2, 7] subarray (elements of the subarray are colored):



Then $K_1 = 3, K_2 = 2, K_3 = 1$, so the power is equal to $3^2 \cdot 1 + 2^2 \cdot 2 + 1^2 \cdot 3 = 20$.

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Yandex.Algorithm 2011: Round 2

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
363752578	Feb/21/2026 01:01	Accepted

→ Problem tags

data structures

implementation

math

two pointers

*2200

No tag edit access

→ Contest materials

Announcement

Tutorial